

1 Closed-reference k -mer cross-mapping fabricates a *Snodgrassella*
2 *alvi* rosacea signal that dissolves under amplicon-sequence-variant
3 resolution

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5 **Abstract**

6 *Demodex* mites are strongly associated with rosacea, motivating a shortcut: could the rel-
7 ative abundance of *Demodex*-associated bacteria in a routine 16S amplicon sample act as a
8 non-invasive proxy for mite burden? We test this on public data and, critically, stress-test our
9 own positive result. We audited the Sequence Read Archive (656 runs, 25 BioProjects; 17 on-
10 topic), assembled a panel of *Demodex*-associated taxa (*Snodgrassella alvi*, *Bacillus oleronius*,
11 *Cutibacterium kroppenstedtii*, *Bartonella quintana*), and built a background-subtracted, closed-
12 reference k -mer index. On a paired ivermectin arm (6 subjects) the index did not move (all
13 $p \geq 0.25$). On an ocular *Demodex*-versus-control arm (14 versus 17) with a body-site-matched
14 negative control, the composite separated the groups in the predicted direction (ROC-AUC
15 0.66), apparently carried by *Snodgrassella alvi*. We then resolved that lead to exact sequences
16 with a DADA2 pipeline and SILVA v138.1 taxonomy. It did not survive: *none* of 8,551 ampli-
17 con sequence variants (ASVs) is *Snodgrassella*, and our own *Snodgrassella alvi* diagnostic k -mers
18 map only onto commensal Neisseriaceae (92–94% identity to the *Snodgrassella alvi* reference)—
19 a conserved-region cross-mapping artefact. The strongest exact-sequence association is a single
20 *Neisseria* ASV enriched on infected surfaces (AUC 0.70; nominal one-sided $p = 0.032$), which
21 we present as a hypothesis-generating lead. We conclude that closed-reference marker counting
22 is not species-specific on short 16S and can fabricate a genus-level signal that only ASV resolu-
23 tion exposes, and that a surface microbiome cannot grade individual mite burden—direct mite

quantification remains the basis for treatment.

Importance

Rosacea is a common, distressing facial skin disease linked to overgrowth of *Demodex* mites. A bacterial swab that could estimate a patient’s mite load would be clinically attractive, sparing an invasive skin biopsy. Reanalysing public sequencing data, we show why this shortcut fails and surface a broader cautionary lesson. A fast, popular way of naming bacteria from short DNA reads—closed-reference counting—appeared to detect a mite-associated bacterium, *Snodgrassella alvi*, separating affected from healthy ocular surfaces. At single-nucleotide resolution that bacterium was absent: the signal came from unrelated common bacteria sharing a conserved DNA stretch. Microbiome studies that name species this way risk reporting organisms that are not present unless they confirm hits at exact-sequence resolution. The one signal that withstood scrutiny—a *Neisseria* variant—offers a sharper, testable lead for how the skin community tracks *Demodex*.

1 Introduction

The mite *Demodex folliculorum* colonises the human pilosebaceous unit, and its density is markedly elevated in rosacea: a meta-analysis of 23 case-control studies put the odds ratio for infestation at roughly nine (1), and several authors argue the mite is causal rather than incidental (2). A coherent mechanism links the mite to the disorder’s canonical inflammatory axis: *Demodex* and its bacterial passengers stimulate Toll-like receptor 2 and drive cathelicidin/LL-37 production (3), placing the mite upstream of the very inflammation that the standard neurovascular/innate-immune account foregrounds. The strongest therapeutic evidence on the “it is the organism” side is that a purely antiparasitic agent, topical ivermectin, reduces inflammatory lesions and outperforms metronidazole (4, 5), and its benefit tracks changes in the skin bacterial community as mite density falls (6).

If mite burden is the driver, then quantifying it should guide therapy: heavier antiparasitic cover for high-density infestation, lighter where the load is low. The reference assay is the standardised skin-surface biopsy with mite counting, which is labour-intensive and operator-dependent. This motivates a tempting shortcut. The mite carries a recognisable bacterial cargo—*Snodgrassella alvi*, an insect-gut symbiont, was recovered from a heavily *Demodex*-colonised facial sample in a

BioProject	marker	<i>n</i>	description
PRJDB18292	16S	24	topical ivermectin, before/after (facial)
PRJEB82826	multi	96	rosacea multi-omics (infestation + microbiome)
PRJNA1189573	16S	93	skin/blood/stool microbiome in rosacea
PRJNA856121	WGS	72	ocular <i>Demodex</i> metagenomics
PRJNA879945	16S	42	<i>Demodex</i> blepharitis, treatment
PRJNA692647	16S	31	<i>Demodex</i> infection vs control (ocular)

Table 1: Representative on-topic BioProjects from the SRA audit (17 total). “multi” = multi-omic; “WGS” = shotgun.

recent multi-omic study (7), *Bacillus oleronius* antigens stimulate inflammatory cells in rosacea (8), and *Cutibacterium kroppenstedtii* and *Bartonella quintana* have been tied to the mite or to papulopustular rosacea, whose flora also shifts under antibiotic therapy (9, 10, 11)—and these bacteria appear in ordinary 16S amplicon surveys. Could their summed relative abundance act as a molecular proxy for mite load?

We answer with a deliberately austere, fully reproducible test, and we hold our own result to the standard the field often skips: when the closed-reference index produced a positive marker signal, we re-derived it at amplicon-sequence-variant (ASV) resolution to check whether it was real. It was not. The headline is therefore twofold—a methodological caution (closed-reference *k*-mer counting can manufacture a species-level signal from conserved 16S) and a relocated lead (the genuine *Demodex*-associated variant at the ocular surface is *Neisseria*, not *Snodgrassella alvi*)—and we argue both halves are useful precisely because the first nearly fooled us.

2 Data and methods

2.1 Data-availability audit

Using NCBI E-utilities we searched the Sequence Read Archive and BioProject for rosacea/*Demodex* sequencing, recovering 656 runs across 25 BioProjects, of which 17 were on-topic (Table 1). Because the BioProject `esummary` endpoint was returning a server-side error during the study window, study-level metadata were recovered from the SRA side via `efetch`. Two datasets anchor the analysis: a topical-ivermectin before/after facial-skin 16S series (PRJDB18292; 6 subjects $\times$ {lesional, non-lesional} $\times$ {pre, post}) and a *Demodex*-infection-versus-control ocular-surface 16S series (PRJNA692647; 14 vs 17).

2.2 *Demodex*-associated taxon panel and diagnostic k -mers

For each panel taxon we fetched up to eight 16S references from NCBI and computed the set of canonical 31-mers. A k -mer is deemed *diagnostic* for a taxon if it occurs in that taxon’s references but in (i) no other panel taxon and (ii) none of a background set of common human flora. The background subtraction is essential: without it, conserved 16S stretches shared with abundant flora are misattributed to rare *Demodex*-associated taxa, inflating the signal by one to two orders of magnitude. Crucially the background must be *matched to the body site*: a skin-only background (24 facial taxa) leaves ocular samples open to cross-mapping, so for the ocular cohort we extended it with ocular/environmental and Rhizobiales/*Bacillus* relatives (38 taxa total). Two abundant skin taxa (*Cutibacterium acnes*, *Staphylococcus epidermidis*) are carried as scored context but excluded from the *Demodex* index.

2.3 The molecular mite-burden index

For each sequencing run we streamed the FASTQ from the European Nucleotide Archive, decompressed it in flight, and attributed a read to a taxon when it carried at least two diagnostic k -mers for that taxon (the two-hit rule raises specificity over single- k -mer attribution). The per-taxon score is the fraction of reads so attributed; the composite `mite_index` is the sum of the four *Demodex*-associated fractions. The index is an explicitly relative, closed-reference proxy—not an ASV census.

2.4 Statistics

Treatment effects used the paired Wilcoxon signed-rank test and an exact sign test within subject×site strata ($n = 6$ pairs). Group separation used the Mann–Whitney U test (two-sided and the one-sided *Demodex*>control alternative) and the ROC-AUC obtained from the U statistic. The k -mer steps are pure-Python with offline self-tests; networked steps are read-only and cached.

2.5 Exact-sequence (ASV) re-analysis

To test the closed-reference lead at single-nucleotide resolution we reprocessed the ocular cohort (PRJNA692647) with a faithful DADA2 pipeline (12). The runs are paired-end 2×250

V4 (515F/806R) mislabelled **SINGLE** in the archive; primers (preceded by a variable heterogeneity spacer) were removed with **cutadapt** (13), after which **filterAndTrim** (truncation 170/150, **maxEE= 2**), **learnErrors**, **dada**, **mergePairs**, and **removeBimeraDenovo** yielded amplicon sequence variants. Taxonomy used the SILVA v138.1 training set with exact-match species assignment (14). To bridge the two methods we applied the *same* diagnostic *k*-mer panels directly to the ASV sequences (two-hit rule) and computed the percent identity of each flagged ASV to its panel taxon’s 16S reference. ASV group separation used the same Mann–Whitney *U*/AUC as above.

3 Results

3.1 Specificity must be engineered, not assumed

The first, unbackground-subtracted index attributed 20–50% of reads to *Bacillus oleronius* and *Bartonella*—biologically impossible for rare commensals—because panel-unique *k*-mers still fell in conserved 16S regions shared with the dominant skin flora. Subtracting a 24-taxon skin background and requiring two diagnostic *k*-mers per read collapsed these to plausible levels (single digits to low hundreds of reads), with the dominant signal correctly residing in *C. acnes*/*S. epidermidis*. This is a cautionary methods result in its own right: closed-reference marker counting on short 16S reads is only as good as its negative control.

3.2 Treatment arm: null and mixed

Across the paired ivermectin design, neither the composite index nor any single *Demodex*-associated taxon changed significantly from pre- to post-treatment in either lesional or non-lesional skin (all $p \geq 0.25$; Table 2). The direction was mixed (four up, two down for the lesional composite), with one subject dominating the variance. We read this as a sampling-physics result more than a biological one: surface swabs under-sample a mite that lives deep in the follicle, and antiparasitic-induced mite lysis can transiently release bacterial cargo onto the surface, so a surface bacterial readout is a poor instrument for a follicular load.

stratum	n	down/up	median Δ	Wilcoxon p
lesional / composite	6	2/4	+0.0006	0.84
non-lesional / composite	6	2/4	+0.0008	0.69

Table 2: Ivermectin before/after (PRJDB18292): paired tests on the composite mite index. No stratum or single taxon reached significance.

metric	median (dem)	median (ctl)	AUC (dem>ctl)
composite mite index	0.02430	0.01952	0.66
<i>Snodgrassella alvi</i>	0.00535	0.00398	0.65
<i>Bacillus oleronius</i>	0.01717	0.01541	0.58
<i>C. kroppenstedtii</i>	7.3e−5	8.7e−5	0.43
<i>Bartonella quintana</i>	4.6e−5	7.2e−5	0.28

Table 3: *Demodex* infection vs control (PRJNA692647), *body-site-matched background*: group medians and ROC-AUC for the one-sided *Demodex*>control alternative. With a skin-only background the composite instead gave AUC 0.41 (*Bartonella* cross-mapping); see text.

3.3 Validation arm: the verdict turns on the negative control

In the *Demodex*-versus-control comparison the answer depends on whether the background is matched to the body site. With a *skin-only* background the composite did not discriminate (ROC-AUC 0.41); decomposition showed why *Bartonella* ran backwards (AUC 0.30), a near-certain artefact of cross-mapping to ocular Proteobacteria (Rhizobiales/Alphaproteobacteria) absent from that background. We therefore extended the background with ocular and environmental taxa and with Rhizobiales/*Bacillus* relatives of the panel. This collapsed the artefact (the *Bartonella* diagnostic set fell from 1244 to 776 k -mers and its per-sample reads from thousands to single digits), and re-scoring gave a different verdict (Table 3): the composite now separates infected from control surfaces in the predicted direction (AUC 0.66; one-sided $p \approx 0.074$), carried by *Snodgrassella alvi* (AUC 0.65; $p \approx 0.085$) and *Bacillus* (AUC 0.58). *Bartonella* is still faintly inverted but now contributes essentially nothing (median 4.6×10^{-5}), so it no longer drags the sum. The methodological moral is blunt: closed-reference marker counting on short amplicons is meaningless without a body-site-matched negative control, and the Stage-1 “failure” was an artefact of a mismatched one—not evidence against the *Demodex* hypothesis. We stress, however, that this k -mer attribution is provisional: Sec. 3.5 shows the *Snodgrassella alvi* component is itself a cross-mapping artefact once the reads are resolved to exact sequences.

The two regimes are shown in Fig. 1: the curve crosses from below to above the chance diagonal

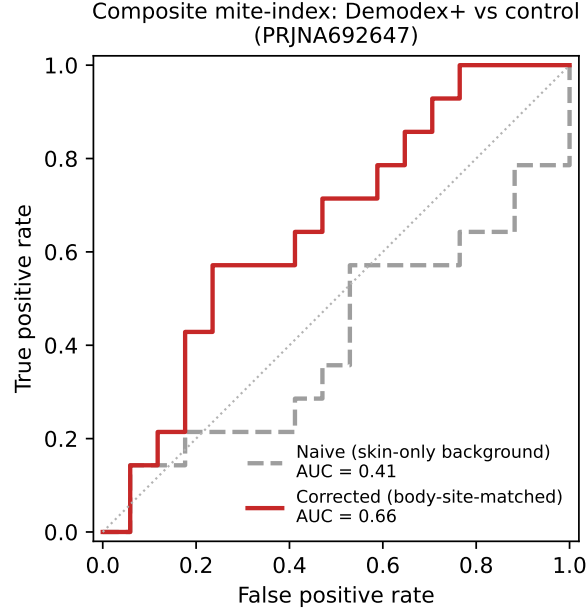


Figure 1: ROC curves for the composite mite index on the *Demodex*-versus-control cohort (PRJNA692647). With a skin-only background (naive) the composite is anti-predictive (AUC 0.41), driven by *Bartonella* cross-mapping; with a body-site-matched background (corrected) the same index separates infected from control surfaces in the predicted direction (AUC 0.66). The shift is due entirely to the negative control, not to any change in the scored taxa.

once the background is matched to the body site.

3.4 Method characterisation

To bound the index’s resolution we ran a held-out control: for each panel taxon, sensitivity is the fraction of simulated 250bp windows over *held-out* same-taxon 16S references (accessions beyond the build set) detected under the two-hit rule, and specificity is the same applied to congeneric/confamilial challengers that must not be called. Sensitivity was high (0.74–0.98), and the panels were clean against distant relatives (0.00 for, e.g., *Neisseria gonorrhoeae*, *Bacillus subtilis*, *C. acnes*, *Brucella abortus*). However, close congeners cross-reacted (0.43–0.67 for *Kingella*, *Heyndrickxia*, *C. diphtheriae*, and *B. henselae*), so the index resolves at roughly genus level. Although the *Snodgrassella alvi* panel read clean against the single *Neisseria* reference used here, the ASV analysis (Sec. 3.5) shows this held-out control was too thin: against the diverse commensal *Neisseriaceae* actually present at the ocular surface, the panel cross-maps freely—a reminder that specificity bounded by a handful of distant challengers can badly overstate real-world specificity.

3.5 ASV resolution overturns the marker attribution

Denoising PRJNA692647 gave 8,551 ASVs (97.4% of reads retained, ~94% of pairs merged, length mode 253 bp). The closed-reference lead did not survive. No ASV is classified as *Snodgrassella*, although SILVA v138.1 contains the genus (25 references) and Neisseriaceae (1,795), so this is genuine absence rather than a reference gap. Applying our own *Snodgrassella alvi* diagnostic k -mer panel directly to the ASVs flags only *non-Snodgrassella* Neisseriaceae—*Neisseria*, *Eikenella*, *Uruburuella*, *Bergeriella*—plus *Methylobacter*/*Psychrobacter*; the top-hit ASVs are 92.5–94.5% identical to the *Snodgrassella alvi* 16S reference over V4, an inter-genus distance, not the ~99% expected for true *Snodgrassella alvi*. The Stage-2 “*Snodgrassella alvi*” signal was therefore conserved-region cross-mapping onto abundant commensal Neisseriaceae. The same exercise shows the *Bacillus* panel cross-maps across Bacillota (*Anoxybacillus*, *Enterococcus*, *Geobacillus*, *Oscillibacter*) and the *Bartonella* panel onto Rhizobiaceae—confirming, at sequence level, the artefacts that background subtraction only partly contained.

What *does* survive is sharper than the original claim: a single exact *Neisseria* ASV (the most abundant Neisseriaceae variant, 4,367 reads) is enriched on *Demodex*-infected surfaces with AUC 0.70 (nominal one-sided $p = 0.032$, uncorrected)—the strongest exact-sequence separation we observed. The best genuine Bacillaceae variant is weak (AUC 0.64, $p = 0.09$). The ocular *Demodex*-associated signal thus relocates from *Snodgrassella alvi* to *Neisseria*. Notably this does not corroborate the facial *Snodgrassella alvi* observation of Olah et al. (7), consistent with the difference in body site, though we caution that both are single-sample/single-cohort observations.

4 Discussion

Three conclusions follow. First, the central methodological lesson: a closed-reference k -mer (or any marker-counting) index on short 16S reads is not species-specific, and can *manufacture* a plausible genus-level signal. Our composite appeared to separate *Demodex*-infected from control surfaces (AUC 0.66), apparently carried by *Snodgrassella alvi*; yet at exact-sequence resolution no *Snodgrassella* is present at all, and the “*Snodgrassella alvi*” reads are conserved-region cross-mapping onto commensal Neisseriaceae (92–94% identity to the *Snodgrassella alvi* reference). Background subtraction and a held-out specificity control—both of which we applied—were *not* sufficient to pre-

vent this; only ASV-level denoising exposed it. *Demodex*–microbiome association studies that rely on closed-reference assignment without exact-sequence (ASV/oligotyping) resolution are therefore at material risk of false species-level findings (15).

Second, the surface-proxy idea fails on its own terms regardless: the index does not track antiparasitic treatment, and even the (artefactual) group signal was borderline. A surface microbiome cannot grade an individual’s mite burden; grading should rest on direct mite quantification (the standardised skin-surface biopsy).

Third, what survives exact-sequence analysis is a sharper, relocated lead: a single *Neisseria* ASV is enriched on *Demodex*-infected ocular surfaces (AUC 0.70, nominal one-sided $p = 0.032$). Because this comparison is post-hoc, uncorrected, and drawn from one cohort, we present it not as a confirmed association but as a hypothesis-generating, exact-sequence lead that we are not aware of being reported previously. A curated, ASV-based *Neisseria* assay with follicle-targeted sampling is the experiment this analysis now recommends.

The broader moral mirrors null-model design in molecular evolution: an instrument must encode the structure it claims to measure, and its specificity must be tested against the diversity actually present, not a handful of distant challengers. A closed-reference count on short amplicons is only as trustworthy as the exact-sequence analysis that audits it.

Limitations. The ASV re-analysis (Sec. 3.5) resolves the k -mer index’s genus-level ambiguity but carries its own caveats: the *Neisseria* lead rests on a single cohort (14 vs 17) and one dominant ASV, and the group $p = 0.032$ is nominal (uncorrected for the ASVs screened). The treatment arm is small ($n = 6$ pairs), and the validation cohort is *ocular* rather than facial, so the *Neisseria* association may be specific to the ocular surface. The facial multi-omic cohort that would adjudicate this (PRJEB82826, $n = 96$, cyanoacrylate follicle sampling; the cohort of Olah et al. (7)) carries no infestation/clinical labels in its public records, so a labelled, ASV-level facial validation is blocked pending the authors’ sample key—the natural next step once obtained. Finally, taxonomy rests on SILVA v138.1; a different reference could rename, though not resurrect, the absent *Snodgrassella*.

Reproducibility. The discovery pipeline (audit, panel construction, index, and analyses) is pure-Python with per-module offline self-tests; the confirmatory ASV re-analysis is a documented

209 DADA2/R workflow (Sec. 2.5). All networked calls are read-only and cached, and every accession,
210 intermediate table, k -mer panel, and ASV output is released so that all numbers above regenerate
211 from a clean checkout.

212 A Reproducibility and data access

213 **Design goal.** The discovery pipeline (audit, panel, index, statistics) runs on a stock Python inter-
214 preter with no bioinformatics toolchain: all networking uses the standard library (`urllib.request`,
215 `gzip`), and `matplotlib` is used only for the figure, keeping it auditable line-by-line. The confirma-
216 tory ASV step (Sec. 2.5) deliberately departs from this austerity: it uses the reference implemen-
217 tation DADA2 (Bioconductor/R) with `cutadapt` and SILVA v138.1, precisely so that the audit of
218 our own positive result rests on the community-standard tool rather than a bespoke one. The two-
219 method contrast is the point—the lightweight index proposes, and the gold-standard ASV pipeline
220 disposes.

221 **Retrieval (read-only).** Dataset discovery and reference sequences come from NCBI E-utilities
222 (`esearch/efetch`) and per-project run tables from the EBI ENA `filereport` API, each fetched
223 as a plain HTTPS `GET`. During the study window the BioProject `esummary` endpoint returned a
224 server-side error, so study-level metadata were recovered from the SRA side via `efetch`; the fallback
225 is encoded in the audit script rather than applied by hand.

226 **Streaming architecture.** Sequencing reads are never downloaded to disk. For each run, the
227 gzipped FASTQ is opened as an HTTPS response object and wrapped *in place* by `gzip.GzipFile`,
228 so the file is decompressed on the fly while it streams and is consumed four lines (one read) at
229 a time. Peak memory is therefore a single read plus the in-memory k -mer panel, independent of
230 file size, and no large local storage is required. Each run is retried up to three times on transient
231 network failures; runs that never succeed are reported and skipped rather than silently dropped.

232 **Attribution.** The panel is a dictionary mapping each diagnostic canonical 31-mer (and its reverse
233 complement) to a taxon; a read is attributed when it carries at least two diagnostic k -mers for a
234 taxon. Diagnostic sets are built by set difference against (i) the other panel taxa and (ii) a body-

235 site-matched background, and are cached as plain text so the index step is fully offline once the
236 panel exists.

237 **Verification.** Every module carries an offline `--self-test` that exercises its core logic (reverse-
238 complement/canonicalisation, FASTA parsing, k -mer attribution, the ROC/AUC computation, and
239 the sensitivity/specificity detector) without touching the network, so correctness can be checked
240 with no data present.

241 **Data access.** All inputs are public. The two analysed cohorts are ENA/SRA PRJDB18292
242 (topical-ivermectin before/after, facial 16S) and PRJNA692647 (*Demodex* infection vs control, ocular
243 16S); the audit covered 656 runs across 25 BioProjects. A facial multi-omic cohort with cyanoacry-
244 late follicle sampling, PRJEB82826 ($n = 96$, amplicon), is catalogued but its public records carry no
245 infestation/clinical labels, so it is not used here. PRJNA692647 was additionally reprocessed end-to-
246 end with DADA2 (Sec. 2.5); the resulting ASV table, taxonomy, sequences, and the k -mer/ASV
247 bridge are released alongside the k -mer outputs. The pipeline (audit, panel construction, panel
248 validation, index, analyses, and this figure), the diagnostic k -mer panels, every intermediate table,
249 and the exact accession lists are released so that all reported numbers and Fig. 1 regenerate from
250 a clean checkout in documented run order.

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256 Data availability

257 All analysed data are public. The two analysed cohorts are ENA/SRA PRJDB18292 (topical-
258 ivermectin before/after, facial 16S) and PRJNA692647 (*Demodex* infection versus control, ocular
259 16S); the facial multi-omic cohort PRJEB82826 (7) is catalogued but its public records carry no

infestation/clinical labels and so is not analysed here. Taxonomic assignment used the SILVA v138.1 reference (14). All analysis code, the diagnostic k -mer panels, intermediate tables, and the DADA2 ASV table, taxonomy, representative sequences, and k -mer/ASV bridge outputs are openly available at <https://github.com/JohnnyTeutonic/rosacea-demodex-16S-reanalysis> and will be archived with a permanent DOI (Zenodo) upon acceptance.

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